

# ULTRON 2.7 Phase 6 Execution Report

Agentic brain layer and safe multi-step task runtime

|           |                                |
|-----------|--------------------------------|
| Project   | ULTRON 2.7                     |
| Phase     | Phase 6: Brain Layer           |
| Workspace | Local workspace checkout: U2.7 |
| Status    | Implemented and verified       |

## Executive Summary

Phase 6 adds ULTRON's brain layer: a task-level orchestration system that can plan multi-step goals, execute allowed steps, pause for confirmation, remember non-sensitive context, and summarize outcomes. The brain does not receive raw shell access.

## Phase 6 Scope

- Add task state models and a TaskPlan structure.
- Add UltronBrain for goal classification, decomposition, preview, execution, and summaries.
- Add short-term session memory and persistent non-sensitive memory.
- Add console and CLI commands for /plan, /do, /memory, and /forget.
- Keep LLM and agentic behavior inside the existing safe runtime boundary.

## What Was Implemented

| Area             | File(s)                                                     | Result                                                              |
|------------------|-------------------------------------------------------------|---------------------------------------------------------------------|
| Brain layer      | src/ultron27/brain.py                                       | TaskState, TaskStep, TaskPlan, MemoryStore, and UltronBrain.        |
| Planner support  | src/ultron27/planner.py                                     | Deterministic append_to_note parsing for multi-step note workflows. |
| Console commands | src/ultron27/console.py                                     | /plan, /do, /memory, and /forget.                                   |
| CLI commands     | src/ultron27/cli.py                                         | Slash commands work in one-shot and interactive modes.              |
| Docs             | README.md, docs/architecture.md, docs/phase6_brain_layer.md | Brain boundary, memory, and command examples.                       |
| Tests            | tests/test_pipeline.py                                      | Expanded to 31 regression tests.                                    |

## Brain Workflow

| Stage       | Purpose                                         | Safety Note                                               |
|-------------|-------------------------------------------------|-----------------------------------------------------------|
| Goal intake | Classify single-step vs multi-step.             | No execution during classification.                       |
| TaskPlan    | Create ordered TaskStep entries.                | Preview uses planner, validator, and policy.              |
| Execution   | Send each step to UltronAssistant.handle().     | Runtime enforces validation, policy, executor, and audit. |
| Pause/stop  | Stop on confirmation, blocked, or failed steps. | High-risk actions cannot silently continue.               |
| Memory      | Record non-sensitive facts and task summaries.  | Passwords, API keys, and tokens are rejected.             |

## Verification Results

| Check              | Command                                                      | Result                                                               |
|--------------------|--------------------------------------------------------------|----------------------------------------------------------------------|
| Unit tests         | <code>python -m unittest discover -s tests</code>            | 31 tests passed.                                                     |
| Acceptance plan    | <code>python -m ultron27 /plan ... --text</code>             | Goal decomposed into create_note and append_to_note.                 |
| Dataset evaluation | <code>python scripts/evaluate_dataset.py --split test</code> | Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0. |
| Safety regression  | <code>python scripts/evaluate_safety_regression.py</code>    | 362 cases, tool accuracy 1.0, policy action accuracy 1.0.            |

## Recommended Next Step

- Move to Phase 7 by connecting this brain runtime to a dedicated visual interface.
- Keep the interface as a client of the brain/runtime API rather than duplicating planner logic.
- Preserve the same confirmation and memory boundaries before adding voice in Phase 8.